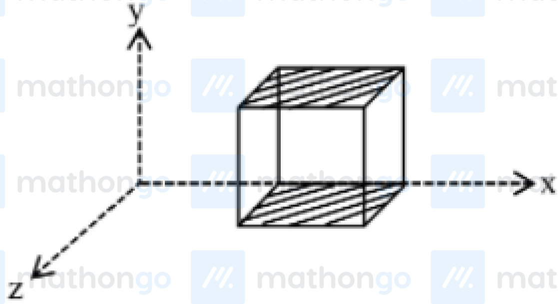
Q1 2021 (01 Sep Shift 2)

A cube is placed inside an electric field, $\overrightarrow{\mathrm{E}}=150 \mathrm{y}^{2} \hat{\mathrm{j}}$. The side of the cube is $0.5 \mathrm{~m}$ and is placed in the field as shown in the given figure. The charge inside the cube is :



(1) $3.8 \times 10^{-11} \mathrm{C}$

(2) $8.3 \times 10^{-11} \mathrm{C}$

(3) $3.8 \times 10^{-12} \mathrm{C}$

(4) $8.3 \times 10^{-12} \mathrm{C}$

Q2 2021 (31 Aug Shift 2)

Choose the incorrect statement :

(a) The electric lines of force entering into a Gaussian surface provide negative flux.

(b) A charge ' q ' is placed at the centre of a cube. The flux through all the faces will be the same.

(c) In a uniform electric field net flux through a closed Gaussian surface containing no net charge, is zero.

(d) When electric field is parallel to a Gaussian surface, it provides a finite non-zero flux.

Choose the most appropriate answer from the options given below

(1) (c) and (d) only

(2) (b) and (d) only

(3) (d) only

(4) (a) and (c) only

Q3 2021 (31 Aug Shift 1)

Two particles A and B having charges $20\mu\mathrm{C}$ and $-5\mu\mathrm{C}$ respectively are held fixed with a separation of 5 cm. At what position a third charged particle should be placed so that it does not experience a net electric force?

$\frac{20\mu\mathrm{C}}{\mathrm{A}} 5\ \mathrm{cm} - 5\mu\mathrm{C}$

(1) At 5 cm from $20\mu\mathrm{C}$ on the left side of system

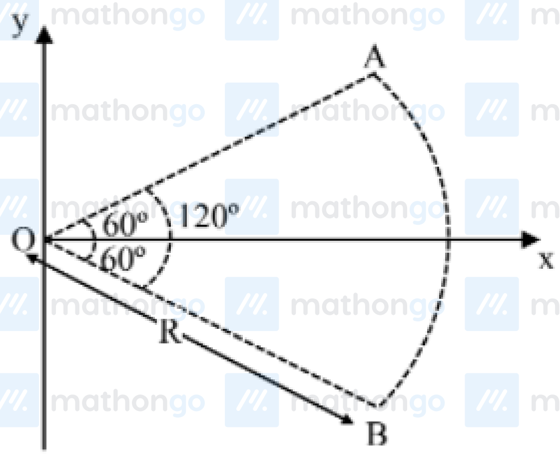
(2) At 5 cm from $-5\mu\mathrm{C}$ on the right side

(3) At 1.25 cm from $-5\mu\mathrm{C}$ between two charges

(4) At midpoint between two charges

Q4 2021 (27 Aug Shift 2)

Figure shows a rod AB, which is bent in a 120° circular arc of radius R. A charge $(-\mathrm{Q})$ is uniformly distributed over rod AB. What is the electric field $\overrightarrow{\mathrm{E}}$ at the centre of curvature O ?



(1) $\frac{3\sqrt{3}Q}{8\pi\varepsilon_0 R^2}(\hat{i})$

(2) $\frac{3\sqrt{3}\mathrm{Q}}{8\pi^2\varepsilon_0\mathrm{R}^2}(\hat{\mathrm{i}})$

(3) $\frac{3\sqrt{3}\mathrm{Q}}{16\pi^2\varepsilon_0\mathrm{R}^2}(\hat{\mathrm{i}})$

(4) $\frac{3\sqrt{3}Q}{8\pi^2\varepsilon_0 R^2}(-\hat{i})$

Q5 2021 (27 Aug Shift 1)

A uniformly charged disc of radius R having surface charge density σ is placed in the xy plane with its center at the origin. Find the electric field intensity along the z-axis at a distance Z from origin :-

(1) $E=\frac{\sigma}{2\varepsilon_0}\left(1-\frac{Z}{\left(Z^2+R^2\right)^{1/2}}\right)$

(2) $E=\frac{\sigma}{2\varepsilon_0}\left(1+\frac{Z}{\left(Z^2+R^2\right)^{1/2}}\right)$

(3) $E=\frac{2\varepsilon_0}{\sigma}\left(\frac{1}{\left(Z^2+R^2\right)^{1/2}}+Z\right)$

(4) $E=\frac{\sigma}{2\varepsilon_0}\left(\frac{1}{\left(Z^2+R^2\right)}+\frac{1}{Z^2}\right)$

Q6 2021 (26 Aug Shift 2)

The two thin coaxial rings, each of radius 'a' and having charges $+Q$ and $-Q$ respectively are separated by a distance of 's'. The potential difference between the centres of the two rings is :

(1) $\frac{Q}{2\pi\varepsilon_0}\left[\frac{1}{a}+\frac{1}{\sqrt{s^2+a^2}}\right]$

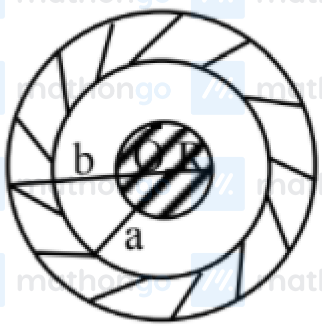
(2) $\frac{Q}{4\pi\varepsilon_0}\left[\frac{1}{a}+\frac{1}{\sqrt{s^2+a^2}}\right]$

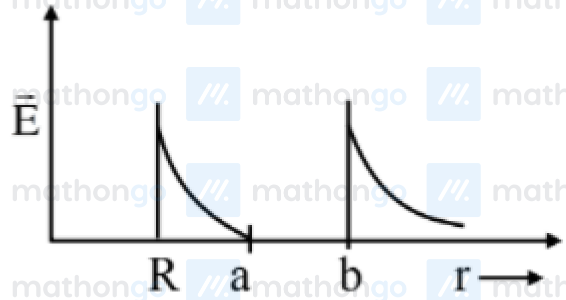
(3) $\frac{Q}{4\pi\varepsilon_0}\left[\frac{1}{a}-\frac{1}{\sqrt{s^2+a^2}}\right]$

(4) $\frac{Q}{2\pi\varepsilon_0}\left[\frac{1}{a}-\frac{1}{\sqrt{s^2+a^2}}\right]$

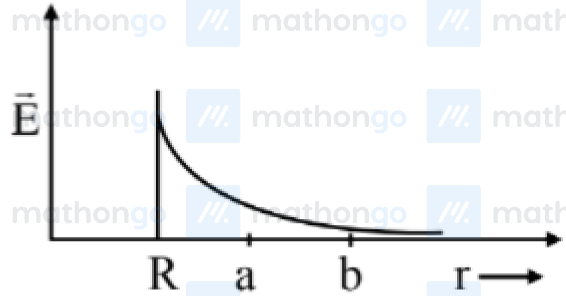
Q7 2021 (26 Aug Shift 1)

A solid metal sphere of radius R having charge q is enclosed inside the concentric spherical shell of inner radius a and outer radius b as shown in figure. The approximate variation electric field $\vec{E}$ as a function of distance r from centre O is given by:

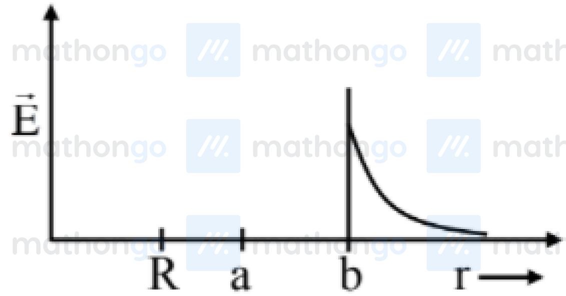




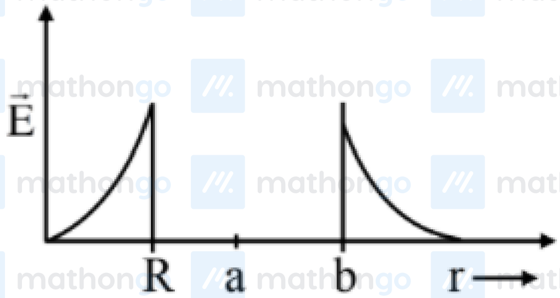
(1)



(2)



(3)



(4)

Answer Key

Q1 (2) **Q2** (3) **Q3** (2) **Q4** (2)

Q5 (1) **Q6** (4) **Q7** (1)

Q1 (2)

As electric field is in y-direction so electric flux is only due to top and bottom surface

Bottom surface $\mathrm{y}=0$

$$\Rightarrow \mathrm{E}=0 \Rightarrow \phi=0$$

Top surface $y=0.5 \mathrm{~m}$

$$\Rightarrow \mathrm{E}=150(.5)^{2}=\frac{150}{4}$$

Now flux $\phi=\mathrm{EA}=\frac{150}{4}(.5)^{2}=\frac{150}{16}$

By Gauss's law $\phi=\frac{Q_{\text{in}}}{\epsilon_{0}}$

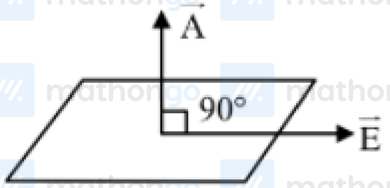
$$\frac{150}{16}=\frac{Q_{in}}{\epsilon_{0}}$$

$$Q_{\text{in}}=\frac{150}{16} \times 8.85 \times 10^{-12}=8.3 \times 10^{-11} \mathrm{C}$$

Option (2)

Q2 (3)

Since $\phi=\overrightarrow{\mathrm{E}} \cdot \overrightarrow{\mathrm{A}}=\mathrm{EA} \cos \theta$



$\theta=90^{\circ}$

$\therefore \phi=0$

Q3 (2)

Null point is possible only right side of -5μC

$$E_N=+\frac{k(-5\mu C)}{x^2}+\frac{k(20\mu C)}{(5+x)^2}=0$$

$$x=5\text{ cm}$$

$\therefore$ option (2) is correct

Q4 (2)

$\varepsilon=\frac{2k\lambda}{\mathrm{R}}\sin\left(\frac{\theta}{2}\right)(-\hat{\mathrm{i}})$

$\lambda=\left(\frac{-\mathrm{Q}}{\mathrm{R}\theta}\right)=\left(\frac{-\mathrm{Q}}{\mathrm{R}\cdot\frac{2\pi}{3}}\right)$

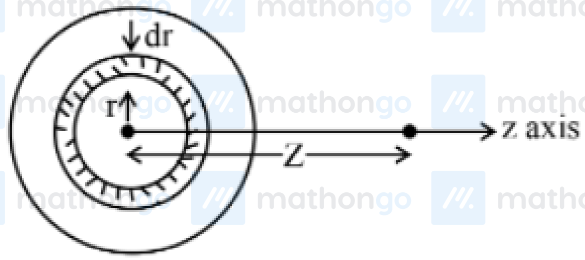
$\lambda=\frac{-3\mathrm{Q}}{2\pi\mathrm{R}}$

$\varepsilon=\frac{2\mathrm{k}}{\mathrm{R}}\cdot\frac{-3\mathrm{Q}}{2\pi\mathrm{R}}\cdot\sin(60^{\circ})(-\hat{\mathrm{i}})$

$\varepsilon=\frac{3\sqrt{3}\mathrm{Q}}{8\pi^2\epsilon_o\mathrm{R}^2}(+\hat{\mathrm{i}})$

Q5 (1)

Consider a small ring of radius r and thickness dr on disc.



area of elemental ring on disc

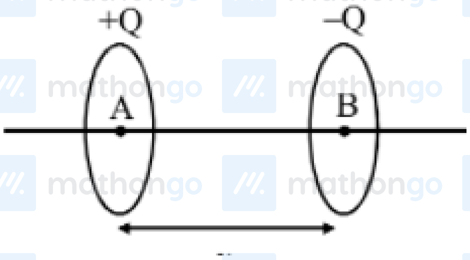
$$dA = 2\pi r dr$$

charge on this ring $dq = \sigma dA$

$$dEz = \frac{kdqz}{\left(z^2 + r^2\right)^{3/2}}$$

$E = \int_0^R dE_z = \frac{\sigma}{2\epsilon_0}\left[1 - \frac{z}{\sqrt{R^2+z^2}}\right]$

Q6 (4)



$$V_A = \frac{KQ}{a} - \frac{KQ}{\sqrt{a^2+s^2}}$$
$$V_B = \frac{-KQ}{a} + \frac{KQ}{\sqrt{a^2+s^2}}$$
$$V_A - V_B = \frac{2KQ}{a} - \frac{2KQ}{\sqrt{a^2+s^2}}$$
$$= \frac{Q}{2\pi\varepsilon_0}\left(\frac{1}{a} - \frac{1}{s^2+a^2}\right)$$

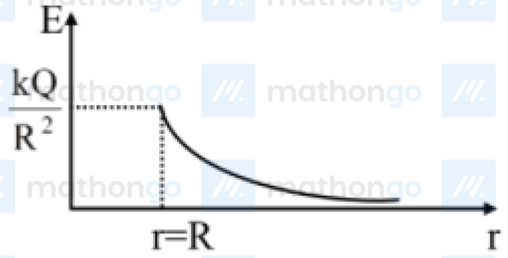
Ans (4)

Q7 (1)

Considering outer spherical shell is nonconducting

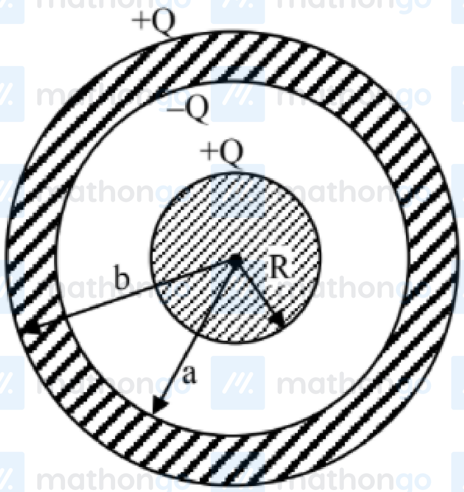
Electric field inside a metal sphere is zero.

$$r < R \Rightarrow E = 0$$
$$r > R \Rightarrow E = \frac{kQ}{r^2}$$



(option 2)

Considering outer spherical shell is conducting



$r < R, E = 0$

$R \leq r < a \qquad E = \frac{kQ}{r^2}$

$a \leq r < b, \qquad E = 0$

$r \geq b \qquad E = \frac{kQ}{r^2}$